

ICO-Cache: Multi-Tiered Semantic Caching and Cost Isolation for LLM Systems

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Abstract

Serving Large Language Models (LLMs) in production is bottlenecked by inference latency and API costs. We present ICO-Cache, an open-source multi-tier caching architecture designed for enterprise RAG and conversational applications. ICO-Cache introduces a strict 7-layer cascade (L0 deterministic, L0b embedding, L1 exact, L2 semantic, L3 context-aware, L4 retrieval, and L5 prompt context) paired with deterministic corpus versioning, complete tenant isolation, and sub-millisecond decision tracing.

1. Multi-Tier Cache Hierarchy

Modern LLM architectures suffer from redundant computation across repeated queries, similar intents, and repeated document retrieval. Existing semantic caches operate solely at the response level, failing when prompt contexts vary slightly. ICO-Cache solves this by implementing an explicit hierarchy spanning every stage of generation:

Layer L0 provides deterministic function caching for static outputs and schemas. Layer L0b caches dense vector embeddings, avoiding repeat calls to embedding models. Layer L1 performs exact hash matching on normalized prompts. Layer L2 applies cosine similarity search in a vector store for semantic equivalence within a configurable threshold. Layer L3 evaluates context-aware caching for conversational threads.

For RAG pipelines, Layer L4 caches retrieved document lists indexed by query hash, top-k, and corpus version. Layer L5 caches assembled prompt contexts indexed by chunk content hashes, preventing duplicate context reconstruction.

2. Corpus Versioning and Cache Isolation

A critical failure mode of RAG caching is stale document reuse: if a document in the corpus is updated or deleted, retrieval caches must never serve invalid historical passages. ICO-Cache solves this through deterministic corpus versioning.

Every chunk in the corpus is fingerprinted by a SHA-256 content hash. The global corpus version is computed as a deterministic digest over all sorted chunk hashes. The L4 retrieval cache key incorporates this corpus version: `l4:{tenant_id}:{query_hash}:{corpus_version}:{top_k}`. If any document in the corpus changes, old corpus cache entries are immediately isolated and never incorrectly reused.

Furthermore, multi-tenant isolation ensures tenant A can never read or overwrite cache entries belonging to tenant B.

3. Decision Tracing and Accounting

To ensure observability and auditability in enterprise deployments, every request entering ICO-Cache produces an immutable DecisionTrace. The decision trace records whether each layer (L0a, L0b, L1, L2, L3, L4, L5, LLM) was attempted, its outcome (HIT, MISS, SKIPPED, BLOCKED), latency in milliseconds, and the exact reuse source.

Concurrently, the Token and Cost Accounting subsystem records input tokens, output tokens, tokens saved, cost, and dollars saved using verified pricing models. Telemetry is streamed in real-time to the production monitoring dashboard.